Lösung: Integrale mit partieller Integration

a) $\int_{-2}^{1} \frac{x}{2} \cdot e^x \, dx$ (v(x), u'(x))

$= \left[\frac{x}{2} \cdot e^x\right]_{-2}^{1} - \int_{-2}^{1} \frac{1}{2} \cdot e^x \, dx$ (v(x), u(x); v'(x), u(x))

$= \left(\frac{e}{2} - \left(\frac{-2}{2} e^{-2}\right)\right) - \frac{1}{2} \int_{-2}^{1} e^x \, dx$

$= \frac{e}{2} + \frac{1}{e^2} - \frac{1}{2}\left[e^x\right]_{-2}^{1}$

$= \frac{e}{2} + \frac{1}{e^2} - \left(\frac{e}{2} - \frac{1}{2e^2}\right) = \frac{1}{e^2} + \frac{1}{2e^2}$

$= 1/e^2 \left(1 + 1/2\right) = \underline{\underline{\frac{3}{2e^2}}}$

b) $\int_{0}^{1} x \cdot (x-3)^5 \, dx$ (u(x), v'(x))

$= \left[x \cdot \frac{(x-3)^6}{6}\right]_{0}^{1} - \int_{0}^{1} 1 \cdot \frac{(x-3)^6}{6} \, dx$ (u(x), v(x); u'(x), v(x))

$= \left(1 \cdot \frac{(1-3)^6}{6} - 0\right) - \int_{0}^{1} \frac{(x-3)^6}{6} \, dx$

$= \frac{2^6}{6} - \left[\frac{(x-3)^7}{7 \cdot 6}\right]_{0}^{1}$

$= \frac{64}{6} - \left(\frac{(-2)^7}{7 \cdot 6} - \frac{(-3)^7}{7 \cdot 6}\right)$

$= \frac{1}{6}\left(64 + \frac{2^7}{7} - \frac{3^7}{7}\right) = \frac{1}{6 \cdot 7}\left(448 + 128 - 2187\right)$

$= -\frac{1611}{42} = \underline{\underline{-\frac{537}{14}}}$

c) $$\int_0^{2\pi} x\cdot\cos(x)\,dx = \Big[x\cdot\sin(x)\Big]_0^{2\pi} - \int_0^{2\pi} 1\cdot\sin(x)\,dx$$

$$= \big(2\pi\cdot\sin(2\pi) - 0\cdot\sin(0)\big) - \int_0^{2\pi}\sin(x)\,dx$$

$$= (0-0) - \Big[-\cos(x)\Big]_0^{2\pi} = \Big[\cos(x)\Big]_0^{2\pi}$$

$$= \cos(2\pi) - \cos(0) = 1-1 = \underline{\underline{0}}$$

d) $$\int_0^1 \overset{u(x)}{x^2}\cdot\overset{v'(x)}{e^{-x}}\,dx = \Big[\overset{u(x)}{x^2}\,\overset{v(x)}{(-e^{-x})}\Big]_0^1 - \int_0^1 \overset{u'(x)}{2x}\,\overset{v(x)}{(-e^{-x})}\,dx$$

$$= \left(\frac{-1}{e} - 0\right) + 2\int_0^1 \overset{u(x)}{x}\cdot\overset{v'(x)}{e^{-x}}\,dx$$

$$= -\frac{1}{e} + 2\cdot\left(\Big[\overset{u(x)}{x}\cdot\overset{v(x)}{(-e^{-x})}\Big]_0^1 - \int_0^1 \overset{u'(x)}{1}\cdot\overset{v(x)}{(-e^{-x})}\,dx\right)$$

$$= -\frac{1}{e} + 2\left(-\frac{1}{e} - 0\right) - 2\int_0^1 (-e^{-x})\,dx$$

$$= -\frac{1}{e} - \frac{2}{e} - 2\Big[e^{-x}\Big]_0^1 = -1/e - 2/e - \left(\frac{2}{e} - 2\right)$$

$$= \underline{\underline{2 - 5/e}}$$

e) $$\int \ln(x)\,dx = \int \overset{v'(x)}{1}\cdot\overset{u(x)}{\ln(x)}\,dx$$

$$= \overset{v(x)}{x}\cdot\overset{u(x)}{\ln(x)} - \int \overset{v(x)}{x}\cdot\overset{u'(x)}{1/x}\,dx$$

$$= x\cdot\ln(x) - \int 1\,dx$$

$$= \underline{\underline{x\cdot\ln(x) - x + C}}$$